	AzureSky		
	Uncertainty Budget	Document no: UB001	
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

AZURESKY (PTY) LTD

MASTER MEASUREMENT UNCERTAINTY BUDGET

(ISO/IEC 17025:2017 Clause 7.6 & SANAS F49 Requirements)

1. Purpose and Scope

This consolidated uncertainty budget encompasses **all calibrated instruments** used in the analytical determination of moisture, ash, volatile matter, fixed carbon, and calorific value.
All calibrations are traceable to **NMISA** through **SANAS-accredited laboratories** (Micron Lab & Montech Calibration Services).

2. The Crux / Nucleus of Method Validation

- **Performance characteristics** (range, accuracy, precision, repeatability) are clearly established.
- **Methods and instruments are validated and verified** for representative coal sample types.
- **Results meet customer and method specifications**, ensuring traceable, reliable measurements.

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3. General Calibration Information

Customer	**AzureSky (Pty) Ltd – Unit 3, 17 Markiel Rd, Wadeville, Germiston 1401**
Contact Person	nonhlanhla@azuresky.it.com
Calibration Providers	Micron Lab (Pty) Ltd – SANAS Accredited. No. 2017/365305/07 Montech Calibration Services (Pty) Ltd – SANAS Accredited
Traceability	All results traceable to NMISA (South Africa)
Confidence Level	Expanded uncertainty with coverage factor *k = 2 (≈ 95 % confidence)*
Decision Rule	Within Tolerance / Out of Tolerance / Indeterminate (as per SANAS F49)

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4. Consolidated List of Calibrated Instruments

Instrument	Manufacturer / Model	Serial No.	Certificate No.	Date of Calibration	Next Due	Expanded Uncertainty (k = 2)	Status
Analytical Balance	BEL M124Ai	ITA2500477	SET 3	25 Mar 2025	Mar 2026	± 0.0004 g	Within Tolerance
Top Pan Balance	AE Adam CKT8H	AEA90242156	AZU T1.1 (CAL 103-03-00 / 14-00)	Mar 2025	Mar 2026	± 0.3 g	Within Tolerance
Platform Scale	Darvan Brenco 150 kg	120330150	SET 5 (25.03.2025)	25 Mar 2025	Mar 2026	± 0.03 kg	Within Tolerance
Furnace (Volatile)	ATS 210325/59V	–	AZU VF1.1 (CAL 103)	Mar 2025	Mar 2026	± 4 °C	Within Tolerance
Furnace (Ashing)	ATS 210325/74A	–	CAL 103	Mar 2025	Mar 2026	± 4 °C	Within Tolerance

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Instrument	**Manufacturer / Model**	**Serial No.**	**Certificate No.**	**Date of Calibration**	**Next Due**	**Expanded Uncertainty (k = 2)**	**Status**
Oven	Darvan Brenco 50 L	DOV53A/210325/40	CAL 103	Mar 2025	Mar 2026	± 2 °C	Within Tolerance
Thermo-Hygrometer	Sinometer SDC103	OBS 202503-01	HI253D13-31 (Montech)	17–18 Mar 2025	18 Mar 2026	± 0.95 % RH / ± 0.2 °C	Within Tolerance
Digital Thermometer + Probe	Sinometer DM6801B	S1208105	OBS T1.1a (Revised)	14 Mar 2025	Mar 2026	± 0.3 °C	Within Tolerance
Weight Set 1	OB WS1 (9 weights, 1 g – 100 g)	–	OB WS1.1R	17 Mar 2025	31 Mar 2026	± 0.00005 – 0.0001 g	Within Tolerance
Weight Set 2	OB WS2 (4 weights, 5 kg – 20 kg)	–	OB WS2.1R	17 Mar 2025	31 Mar 2026	± 0.09 – 0.5 g	Within Tolerance

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5. Uncertainty Budget Summaries

5.1 Balances and Weights

Balances and weights are evaluated jointly since mass calibration forms a single chain of traceability.

Source	Estimate	Distribution	Divisor	u(xi)	Type	Notes
Calibration (Weights + Balance)	0.0004 g – 0.3 g	Normal	2	0.0002 – 0.15	B	From SANAS certificates OB WS1.1R & WS2.1R
Repeatability	0.0001 g – 0.1 g	Normal	1	0.0001 – 0.1	A	10 replicates
Resolution	Half readability	Rectangular	√3	0.00006 – 0.06	B	Instrument display
Air Buoyancy Effect	Negligible	Rectangular	√3	–	B	For small weights only

Expanded U (k = 2):

- Analytical Balance = ± 0.0005 g
- Top Pan Balance = ± 0.4 g
- Platform Scale = ± 0.04 kg

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- Weights = ± 0.0001 g to ± 0.5 g

5.2 Temperature-Controlled Equipment

(Ovens and Furnaces, including Digital Thermometer Probe)

Source	Estimate	Distribution	Divisor	u(xi)	Type	Notes
Controller Resolution	1 °C	Rectangular	√3	0.58	B	Display limit
Sensor Drift	1 °C	Rectangular	√3	0.58	B	Thermocouple stability
Mapping Variation	1 °C	Normal	1	1	A	Temp. mapping
Calibration Certificate	2–4 °C	Normal	2	1–2	B	SANAS certificates CAL 103 / OBS T1.1a

Expanded U (k = 2):

- Oven = ± 3 °C Furnaces = ± 5 °C Thermometer = ± 0.3 °C

5.3 Thermo-Hygrometer

Parameter	Source	Estimate	Distribution	u(xi)	Type	Notes
Temperature (°C)	Certificate HI253D13-31	± 0.2 °C	Normal	0.1 °C	B	At 22.5 °C, 47 % RH
Humidity (% RH)	Certificate HI253D13-31	± 0.95 % RH	Normal	0.48 % RH	B	Range 15 – 77 % RH
Repeatability	0.1 °C / 0.2 % RH	Normal	1	0.1 °C / 0.2 % RH	A	From replicates under lab conditions

Expanded U (k = 2): ± 0.2 °C and ± 0.95 % RH

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6. Combined Analytical Uncertainty

Propagation of the individual equipment uncertainties shows that total analytical uncertainty remains well below the precision limits of ISO methods 589, 1171, 562, 5925 and ISO 1928.
Therefore, measurement uncertainty does not materially affect analytical results, confirming fitness for purpose.

7. Traceability and Compliance

All instruments are traceable to national standards through SANAS-accredited calibration.
Each certificate explicitly references **ISO/IEC 17025** compliance, ensuring an **unbroken chain of traceability** to the **SI units**.

8. Ash Specific Uncertainty budget

Contributor	Estimate / Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
Method Repeatability (s)	0.2356%	Normal	1	0.2356%	**A**	**AS003MV Section: Repeatability**
Intermediate Precision	0.2942%	Normal	1	0.2942%	**A**	**AS003MV Section: Precision**

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Contributor	Estimate / Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
CRM Uncertainty (AS002)	0.135%	Normal	2	0.0675%	**B**	**AS003MV Section: Uncertainty**
Balance Calibration	0.00015g	Normal	2	0.000075g	**B**	**UB001 Section 4 / AS003MV**
Furnace Calibration	4.0°C	Normal	2	2.0°C	**B**	**UB001 Section 5.2 / AS003MV**

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9. Calorific Value Specific Uncertainty budget

Contributor	Estimate / Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
Balance Calibration	0.0004g	Normal	2	0.0002g	B	**UB001 Sec 4** (Analytical Balance)
Digital Thermometer	0.3°C	Normal	2	0.15°C	B	**UB001 Sec 4** (Probe Calibration)
Method Repeatability	0.0144MJ/kg	Normal	1	0.0144MJ/kg	**A**	**AS002MV Sec 3.4.1** (Validation Data)
CRM Homogeneity	0.0117MJ/kg	Normal	1	0.0117 MJ/kg	**A**	**AS002MV Sec 3.4.1** (In-house CRM)
CRM Calibration/Value	0.1083 MJ/kg	Normal	2	0.0542MJ/kg	**B**	**AS002MV Sec 3.4.1** (In-house CRM)
Bomb Stability	Negligible	Rectangular		-	B	**AS002MV Sec 3.4.1**

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10. ISO Volatile Specific Uncertainty budget

Contributor	Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
Method Repeatability	0.417	Normal	1	0.417%	**A**	**AS001MV Section 4**
Balance Calibration	0.0004g	Normal	2	0.0002g	B	**UB001 Section 4**
Furnace Temperature	4 °C	Normal	2	2.0°C	B	**UB001 Section 5.2**
Robustness (Time)	1 min	Rectangular	3	0.577 min	B	**AS001MV Section 9**

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11. ISO Moisture Specific Uncertainty budget

Contributor	Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
Repeatability (sr)	0.03518%	Normal	1	0.03518%	**A**	**AS004MV Section 5**
Reproducibility (R)	0.21374%	Normal	1	0.2137%	**A**	**AS004MV Section 4**
Oven Temperature	2 °C	Normal	2	1.0 °C	B	**UB001 Section 5.2**
CRM Bias (CRM002)	0.0338%	Normal	1	0.0338%	B	**AS004MV Section 8**

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8. Conclusion

The integrated uncertainty budget demonstrates that:

- All instruments are within tolerance, fit for use, and traceable.
- The combined expanded uncertainty at 95 % confidence satisfies ISO/IEC 17025 Clause 7.6.
- The laboratory maintains a single, structured **uncertainty framework** suitable for continuous expansion as new equipment is introduced.

11. Revision

Revision no	Content changed	Reason for revision
01	No content changed, but rather an addition	The procedure is revised to include detailed uncertainty budget tables for Volatile Matter, Inherent Moisture, Ash, and Calorific Value. Each table now explicitly lists both Type A and Type B contributors with their respective data sources

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